Cluster Profile Report

Customer Segmentation using RFM Analysis and Clustering

Executive Summary

This project segments customers based on their purchasing behavior using RFM (Recency, Frequency, Monetary) analysis and clustering techniques. The objective is to identify groups of customers with similar behaviors and provide actionable marketing recommendations for each segment.

The analysis was performed on an e-commerce transaction dataset. Customer-level features were engineered using RFM metrics and clustered using K-Means and DBSCAN algorithms.

Methodology

RFM Analysis

Three customer behavior metrics were calculated:

Recency (R)

Number of days since the customer's most recent purchase.

Frequency (F)

Number of unique transactions made by the customer.

Monetary (M)

Total amount spent by the customer.

After feature engineering, logarithmic transformation and standardization were applied to prepare the data for clustering.

Clustering Approach

K-Means Clustering

- Optimal number of clusters determined using Elbow Method.
- Cluster quality evaluated using Silhouette Score.
- Generated interpretable customer segments suitable for business decision-making.

DBSCAN Clustering

- Used to identify dense customer groups and outliers.
- Useful for anomaly detection but less interpretable for marketing segmentation.

Cluster Profiles

Cluster 0 – Champions

Characteristics

- Lowest recency values
- Highest purchase frequency
- Highest monetary contribution

Business Interpretation

These are the most valuable customers who purchase frequently, spend heavily, and have purchased recently.

Recommended Actions

- VIP membership programs
- Exclusive product launches
- Personalized offers
- Premium customer support
- Loyalty rewards

Cluster 1 – At-Risk / Lost Customers

Characteristics

- High recency values
- Low purchase frequency
- Low spending

Business Interpretation

Customers who have not purchased recently and show signs of churn.

Recommended Actions

- Win-back email campaigns
- Discount coupons
- Personalized reminders
- Limited-time promotional offers

Cluster 2 – Loyal Customers

Characteristics

- Moderate recency
- High frequency
- Moderate to high spending

Business Interpretation

Customers who regularly purchase and contribute consistently to revenue.

Recommended Actions

- Product recommendations
- Loyalty point programs
- Cross-selling campaigns
- Subscription offers

Cluster 3 – Big Spenders

Characteristics

- High monetary value
- Lower purchase frequency
- Large average order value

Business Interpretation

Customers who spend significant amounts but purchase less frequently.

Recommended Actions

- Premium product bundles
- Personalized marketing campaigns
- Early access to premium collections
- Upselling opportunities

Key Insights

1. Customer purchasing behavior varies significantly across segments.
2. A small group of customers contributes a disproportionately large share of revenue.
3. High-value customers should receive personalized retention strategies.
4. At-risk customers require re-engagement campaigns to reduce churn.
5. Segment-specific marketing can improve conversion rates and customer lifetime value.

Model Evaluation

Elbow Method

Used to identify the optimal number of clusters by analyzing Within-Cluster Sum of Squares (WCSS).

Silhouette Score

Measured cluster separation and cohesion to validate clustering performance.

DBSCAN Findings

DBSCAN successfully identified outliers and dense customer groups but produced less interpretable business segments compared to K-Means.

Conclusion

The customer segmentation framework successfully transformed raw transaction data into meaningful customer groups. The results demonstrate how RFM analysis combined with clustering can support targeted marketing, improve customer retention, and increase overall business value.

K-Means clustering provided the most actionable customer segments and is recommended for deployment in customer relationship management and marketing personalization initiatives.